Model Validation, Overfitting Control & Hyperparameter Tuning

Artificial Intelligence & Machine Learning Internship — Task 3 Technical Report

Domain: Machine Learning & Data Science

Dataset: California Housing Dataset

Evaluation Focus: Cross-Validation, GridSearchCV, Generalization

Status: Completed & Verified

1. Objective & Executive Summary

Building upon the regression workflow established in Task 2, this project advances into professional model reliability, generalization assessment, and systematic tuning. While baseline modeling focuses on fitting algorithms to training data, production-grade Machine Learning demands models that perform consistently on unseen real-world data.

The main focus of Task 3 is identifying and mitigating **Overfitting**, moving beyond simple train-test splits via **5-Fold Cross-Validation**, and optimizing parameters using **GridSearchCV**. Evaluation was carried out using the standard California Housing dataset, comparing original baseline regression algorithms with a fine-tuned Decision Tree Regressor.

2. Overfitting & Underfitting Analysis

In initial trials, an unconstrained Decision Tree Regressor was fitted to the training data. Tree-based models tend to memorize training instances down to individual leaf nodes if left unconstrained, leading to high variance (Overfitting).

UNCONSTRAINED TRAIN RMSE

~0.000

Near-zero training error (memorization).

UNCONSTRAINED TEST RMSE

~0.735

Significant error gap on test data.

TUNED MODEL TEST RMSE

0.645

Substantial error reduction post-tuning.

Key Insights: The large discrepancy between the unconstrained train error and test error confirmed severe overfitting. By limiting tree complexity (restricting maximum depth and requiring minimum samples for splits), we effectively bounded model variance, improving generalization performance.

3. Validation & Hyperparameter Tuning Methodology

To prevent optimistic evaluation bias from a single train-test split, a **5-Fold Cross-Validation** approach (`cv=5`) was implemented. The data was split into five equal partitions, iteratively training on four folds and validating on the remaining fold.

Hyperparameter Search Space (GridSearchCV):
Using scikit-learn's `GridSearchCV`, an exhaustive search was performed across key decision tree parameters to balance bias and variance:

- `max_depth` : Candidate values `[3, 5, 7, 10]` to constrain tree depth.
- `min_samples_split` : Candidate values `[2, 5, 10]` to enforce minimum sample requirements for splitting node branches.

The optimal configuration identified via cross-validation proved to control node expansion while retaining essential nonlinear feature interactions.

4. Model Performance Comparison & Final Justification

The tuned Decision Tree Regressor was evaluated against baseline Linear Regression and Ridge Regression models established in Task 2. All models were assessed using Root Mean Squared Error (RMSE) and Coefficient of Determination (R^2 Score).

Model Architecture	RMSE (Lower is Better)	R^2 Score (Higher is Better)	Status
Linear Regression (Baseline Task 2)	0.745581	0.575788	Baseline
Ridge Regression (Regularized Task 2)	0.745554	0.575819	Linear Optimal
Tuned Decision Tree (Task 3 GridSearch)	**0.645430**	**0.682099**	**Selected Best**

Scientific Justification for Final Selection:

- **Accuracy & Error Reduction:** The Tuned Decision Tree achieved an RMSE of **0.6454** compared to ~0.7455 for linear models, representing a ~13.4% reduction in prediction error.
- **Explanatory Power:** The R^2 score improved significantly to **0.6821** (explaining over 68.2% of target variance), outperforming linear models (~57.5%).
- **Nonlinear Modeling Capability:** Housing price dynamics involve complex non-linear spatial interactions (e.g., location features) that decision trees capture more effectively than linear equations.
- **Controlled Variance:** Hyperparameter tuning prevented the tree from overfitting, ensuring reliable performance on unseen holdout test data.

5. Summary & Key Learnings

Task 3 successfully demonstrated the practical necessity of hyperparameter tuning and robust model validation. By combining 5-fold cross-validation with systematic grid search, the resulting Decision Tree Regressor delivers strong generalization capabilities, low error rates, and stable predictive performance ready for production workflows.